Review Article

Self-Acupressure for Symptom Management in Cancer Patients: A Systematic Review

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Abstract

Context. Acupressure is a popular nonpharmacological intervention that is increasingly proven to effectively alleviate symptoms in patients with cancer. However, the effects of self-acupressure on cancer symptom management are less clear.

Objectives. This systematic review is the first to summarize the current experimental evidence on self-acupressure for symptom management in cancer patients.

Methods. Eight electronic databases were searched for experimental studies that examined self-acupressure for cancer patients with symptoms and published in peer-reviewed English or Chinese journals. The methodological quality of the included studies was evaluated using the revised Cochrane risk-of-bias assessment tool and the JBI critical appraisal checklist for quasi-experimental studies. Data were extracted as predefined and synthesized narratively. The Template for Intervention Description and Replication checklist was used to report the intervention characteristics.

Results. A total of 11 studies were included in this study, six as feasibility or pilot trials. The methodological quality of included studies was suboptimal. Substantial heterogeneity was observed in acupressure training, acupoint selection, intervention duration, dosage, and timing. Self-acupressure was only associated with reduced nausea and vomiting ($P = 0.006$ and $P = 0.001$).

Conclusion. The limited evidence from this review precludes the definitive conclusions on intervention effectiveness for cancer symptoms. Future research should consider developing the standard protocol for intervention delivery, improving the methodology of self-acupressure trials, and conducting large-scale research to advance the science of self-acupressure for cancer symptom management. J Pain Symptom Manage 2023;66:e109−e128. *© 2023 American Academy of Hospice and Palliative Medicine. Published by Elsevier Inc. All rights reserved.*

Key Words

Cancer, symptom, acupressure, systematic review, nausea, vomiting

Introduction

Symptom management is a central practice in cancer care. Although pharmacological treatments remain the major modality for symptom management, nonpharmacological interventions are becoming favored by cancer patients due to quality of life (QOL) benefits and having limited adverse effects.[1,2] Acupressure is a popular nonpharmacological intervention that has increasingly been proven to effectively alleviate cancer-related symptoms, including pain, fatigue, insomnia, nausea, and vomiting and improve QOL.[3-7] It is defined as a traditional Chinese medicine (TCM) treatment to stimulate acupoints (*Qi* hole) along the meridians (*Qi* channels) in the body by applying light to moderate physical pressure with fingers by a trained therapist.[8] According to the WHO's standard acupoint nomenclature, there are 361 acupoints located on 12 primary meridians.[9] Applying acupressure is theorized to remove stagnant *Qi* as the root of symptoms and restore its natural flow, subsequently reducing symptoms.[3,10,11]

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Because acupressure is easy to learn by patients, it has been gradually developed as a patientself-administered therapy for chronic pain and insomnia.[12,13] Compared with acupressure administered by a therapist, self-acupressure is more economical and convenient because patients can make self-practice at any time after minimal training[13-15] Despite the practical advantages of self-acupressure, little is known about its effectiveness, particularly on cancer symptoms. Only one published systematic review has addressed this intervention thus far.[14] However, this review included eligible papers published until 2013 and targeted patients with different health conditions. Only two studies examined cancer patients, and the results showed the potential of self-acupressure in reducing chemotherapy-induced nausea and vomiting.[16,17] Furthermore, the intervention components of self-acupressure trials for symptom management have yet to be evaluated, thereby impeding the studies from being replicated. Additionally, this review failed to provide evidence on the effectiveness of self-acupressure on QOL, which is a core patient-reported outcome affected by symptoms.[18]

To our knowledge, no systematic review regarding self-acupressure for cancer symptom management has been published. Understanding the potential effects of self-acupressure on cancer-related symptoms and QOL and its key intervention components would provide a useful starting point to inform future intervention development, testing, and implementation. Hence, this systematic review was the first study to summarize the current evidence on self-acupressure for symptom management in cancer patients. Specifically, the objectives included 1) examining the effects of self-acupressure on symptoms and QOL and 2) characterizing the key features of self-acupressure intervention protocol.

Materials and Methods

This systematic review was reported following the updated Preferred Reporting Items for Systematic Reviews and Meta-Analyses Checklist (Appendix I).

Eligibility Criteria

The population, intervention, comparator, outcome, and study design (PICOS) framework was used to formulate the inclusion and exclusion criteria. Studies were included if they 1) examined cancer patients with any symptoms caused by cancer and its treatments; 2) used self-acupressure as the intervention; 3) included usual care, sham intervention, active control, and waitlist control as the comparator or no comparator; 4) measured at least one symptom as the primary outcome and/or QOL as the secondary outcome using patient-reported outcome measures;[18] and 5) were randomized controlled trials (RCTs), quasi-experimental, or pre-post experimental studies. Only peer-reviewed studies published in English or Chinese language journals were included. Studies that examined auricular acupressure were excluded because ear acupoints are related to all parts of body organs but may or may not directly connect to meridians. In contrast, body acupressure is based on acupoint activation along the meridians.[19] We also excluded studies that contained acupressure bands without applying pressure by patients themselves or a combination of self-acupressure and other TCM therapies.

Information Sources and Search Strategy

Eight electronic databases were searched including PubMed, CINAHL, British Nursing Index, Cochrane Central Register of Controlled Trials, Embase, Web of Science, China National Knowledge Infrastructure, and Index to Taiwan Periodical Literature System. Due to the nature of the first systematic review, the search period was set from the inception of each database to 29 January 2022 to identify as many relevant studies as possible. The keywords included "cancer" and "self-acupressure" (Appendix II). The search strategies for each database are shown in Appendix III. The literature search was supplemented by manual searching of the reference lists of the included studies and published systematic reviews.

Selection Process

The retrieved records were imported into Endnote 20.0. Following the removal of duplicates, two researchers screened the title and abstract of each paper independently and excluded those that did not meet the eligibility criteria. Potentially eligible records were examined via full-text assessment, and consensus was reached after discussion.

Data Extraction

Following a predefined data extraction form,[20] data regarding authors, country, publication year, study design, participant characteristics, intervention group, control group, outcome assessment, and assessment time points, as well as outcome data, were independently extracted from each eligible study by the two researchers. For intervention characteristics, data extraction was guided by the Template for Intervention Description and Replication (TIDieR) checklist.[21] which included the rationale (why), what materials were used, what procedures, who provided the intervention, how, where, when and how much the intervention was delivered, any intervention tailoring and modifications, and how well the intervention was delivered. Acupoint selections were extracted according to the WHO Standard Acupuncture Point Locations in the Western Pacific Region.[9] Extracted data were cross-checked, and any disagreements were resolved through discussion.

Risk of Bias Assessment

The revised Cochrane risk-of-bias tool (RoB 2.0) for randomized trials was used to assess the methodological quality of all studies except for one quasi-experimental study.[22] The nine-item JBI critical appraisal checklist for quasi-experimental studies was used for one quasi-experimental study to evaluate its methodological quality.[23] All studies were appraised by the two researchers, and any disagreements were resolved by group discussion.

Summary of Results

A meta-analysis was planned for this review but was not performed because of the substantial variations in self-acupressure interventions, control groups, and outcome measurement tools. Hence, the predefined data from each included study were described narratively.

Results

Study Selection

Database searching yielded 372 records, while five were derived from manual searching (Fig. 1). After removing duplicates ($n = 78$), the titles and abstracts of 294 records were reviewed, and 263 records were excluded, leaving 31 records for full-text review. Twenty-two papers were excluded due to interventions other than self-acupressure, conference abstracts and protocol paper. Ultimately, 11 papers were eligible and retained for this review.[16,17,24-32]

Study Characteristics

The eleven studies were published between 2004 and 2021 (Table 1), and were conducted in the United States ($n = 3$),[17,25,28] the United Kingdom ($n = 2$),[24,27] Korea ($n = 2$),[16,32] Taiwan ($n = 1$),[26] Hong Kong (n = 1),[29] Turkey ($n = 1$),[30] and Vietnam ($n = 1$).[31] Among the 11 studies, four studies were definitive RCTs,[17,28,30,32] six were feasibility or pilot RCTs,[24–27,29,31] and one was quasi-experimental.[16] The seven studies comprised three arms,[17,24-28,31] with two arms in four studies.[16,29,30,32] For the control group, usual or enhanced usual care was applied in six studies,[16,17,27,28,30,31] followed by the sham group in four studies,[17,24,26,31] and active control (health education and acupressure with different intensities) in four studies.[25,28,29,32] Sample sizes ranged from seven[27] to 288.[28]

Risk of Bias in Studies

The risk of bias assessment was performed for a quasi-experimental study and ten RCTs. The quasi-experimental study scored "yes" for all items except for complete follow-up and statistical analysis. Using RoB 2.0 (Table 2), nine of 10 RCTs were rated as "high risk of bias",[17,24,26,27,30,32] or "some concerns."[25,28,31] Reasons for poor quality stemmed from limited to no

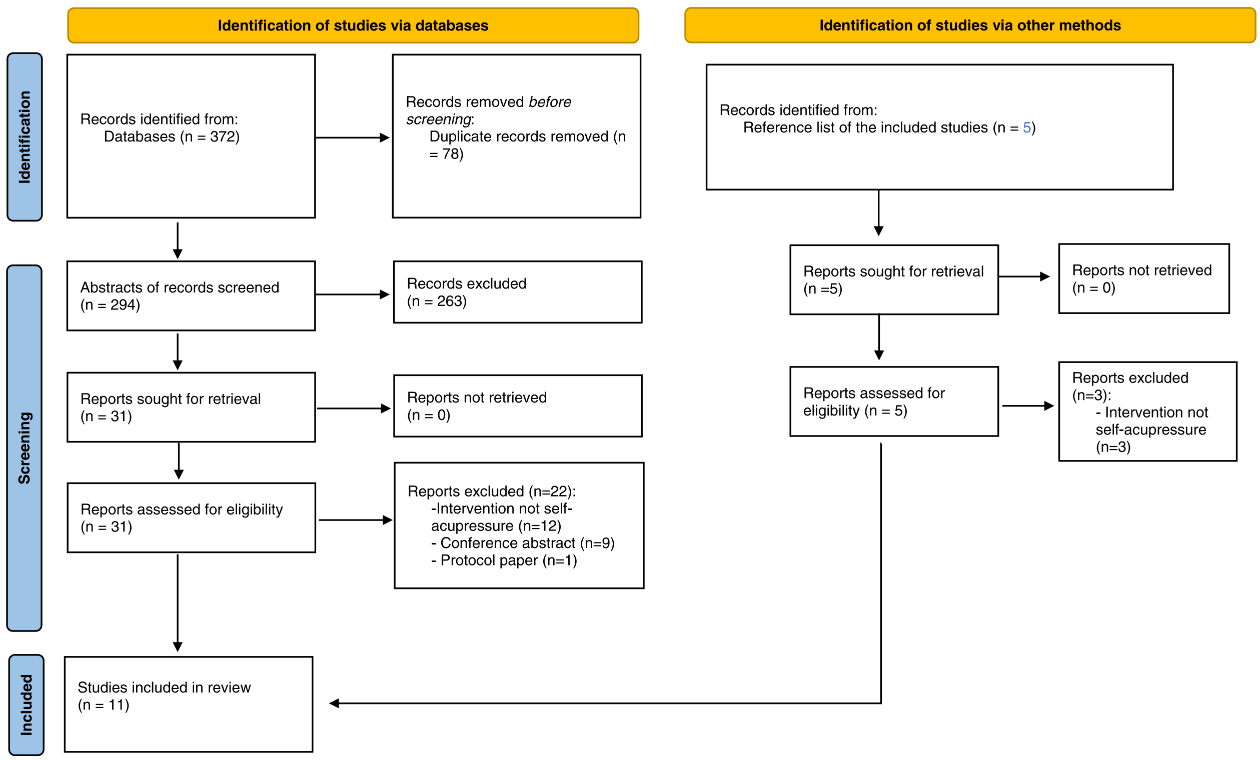

Fig. 1. PRISMA 2020 flow diagram.

Table 1
Characteristics of Self-Acupressure Intervention for Cancer Symptom Management (N = 11)

No.	Author/ Country/ Pub Year	Study Aim	Study Design	Participants	Intervention Group	Control Group	Outcome Measures and Assessment Time Points	Outcome Data
1	Shin et al. (2004)[16] South Korea	To examine the effect of acupressure on emesis control in postoperative gastric cancer patients undergoing chemotherapy	Quasi-experimental	N = 40 (I = 20, C = 20), gastric cancer patients during chemotherapy Mean age = 50 yrs, female = 30%, stage II-IV, time since diagnosis not mentioned	Self-acupressure	Usual care	Baseline (Day one), postintervention (Day two to five) Nausea & vomiting (nausea subscale of Rhode's Index of Nausea, Vomiting, and Retching)	The average severity of nausea and vomiting in intervention and control groups from day one to five were 4.35 ± 2.75 and 7.95 ± 5.91, respectively, with a difference of 3.6 ($P = 0.001$)
2	Dibble et al. (2007)[17] USA	To compare differences in chemotherapy induced nausea and vomiting among three groups of women (Acupressure, placebo acupressure, and usual care) undergoing chemotherapy for breast cancer	RCT	N = 160 (I = 53, C1 = 53; C2 = 54), breast cancer patients during chemotherapy who reported moderate nausea (≥3 on the worst nausea item by the Morrow Assessment of Nausea and Emesis) Mean age = 49.3 yrs, female = 100%, time since diagnosis and cancer staging not mentioned	Self-acupressure	C1 = Sham acupressure C2 = Usual care	Baseline (Day 1), postintervention (Day two to 11) Nausea & vomiting (nausea subscale of the Rhodes Index of Nausea).	No significant difference was found in the intensity of acute nausea and vomiting by treatment groups ($P = 0.547$). The acupressure group had a statistically significant reduction in the intensity of nausea and vomiting ($P = 0.006$) when compared with the sham acupressure and usual care groups.
3	Molassiotis et al. (2007)[24] UK	To assess effects of acupuncture and acupressure in managing cancer-related fatigue and assess feasibility of running a randomized trial with these two complementary therapies in preparation for a large trial	Feasibility RCT	N = 47 (IA = 15, IB = 16, C = 16), cancer survivors who completed chemotherapy ≥ 1month and reported a high level of fatigue (>5 on a 0−10 Likert scale) Mean age = 53.4 yrs, female = 68.1%, time since diagnosis and cancer stage not mentioned	IA: Acupuncture IB: Self-acupressure	Sham acupressure	Baseline (before randomization), postintervention (week two), two weeks postintervention Fatigue (MFI) Adverse effects (self-reported)	Both acupuncture and self-acupressure were more significantly effective than sham group in reducing fatigue, prominently for general fatigue ($P = 0.024$ to <0.001). The reduction in general fatigue for acupuncture, acupressure, and sham control from baseline to postintervention were 5.9, 3.2, and 0.1 respectively. The significant between-group differences remained at two weeks postintervention (*P* value not provided) Adverse effect: only one patient reported bruising from pressure and pain in the point after pressure.
4	Zick et al. (2011)[25] USA	To compare the effects of high- and low doses of stimulating acupressure, as compared to relaxation acupressure on severity of fatigue and evaluate safety, tolerability, adherence, blinding, and beliefs/expectation of	Pilot RCT	N = 43 (I = 14, C1 = 14, C2 = 15), cancer survivors ≥12 weeks post-treatment and reported moderate to severe fatigue (≥4 on BFI) Mean age = 54.0 yrs, Female = 97.7%, time since diagnosis = 37.9 to	1. Relaxation acupressure	C1: Low-dose stimulatory acupressure C2. High-dose stimulatory acupressure	Baseline, postintervention (week 12) Fatigue (BFI) Safety & tolerability (CTCAE 3.0)	The severity of fatigue significantly reduced across all treatment groups, with significantly greater reductions in the relaxation group (4.0 ± 1.5) than high- (2.2 ± 1.6) and low-dose (2.7 ± 2.2) stimulatory acupressure groups at postintervention ($P = 0.027$).

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Table 1
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No.	Author/ Country/ Pub Year	Study Aim	Study Design	Participants	Intervention Group	Control Group	Outcome Measures and Assessment Time Points	Outcome Data
		participants of the three acupressure treatments		44.6months, stage I-IV (stage I = 58.14%)				There was no difference between the groups in terms of total adverse events (P = 0.45−0.64). Only ≤grade one adverse effects were reported in nine cases, including musculoskeletal, dizziness, hot flashes, and sleep issues.
5	Tang et al. (2014)[26] Taiwan	To explore the effects of self-acupressure on fatigue of lung cancer patients undergoing chemotherapy	Pilot RCT	N = 57 (IA = 17, IB = 24, C = 26), lung cancer patients receiving chemotherapy Mean age = 58.3 yrs, female = 42.1%, stage I-IV (stage IV 66.7%), time since diagnosis not mentioned	IA: Self-acupressure with essential oils IB. Self-acupressure only	Sham acupressure	Baseline (one day before the first chemotherapy), Day one of the third chemotherapy, postintervention (Day one of the 6th chemotherapy) Fatigue (Tang Fatigue Rating scale) Sleep (PSQI) Anxiety and depression (HADS)	The between-group change in fatigue between self-acupressure with essential oils, self-acurpessure and sham acupressure was not significant, with a mean difference of 21.90 (P = 0.57) and 17.66 (P = 0.59), respectively. For sleep quality, a significant difference of 2.25 was identified between self-acupressure and sham acupressure at postintervention (P = 0.04). For anxiety, the mean differences in the changes between self-acupressure with essential oils and sham acupressure and between self-acupressure and sham acupressure from baseline to post intervention were -1.58 and − 1.02, respectively, but results were not statistically significant (P = 0.289, P = 0.656). For depression, the differences in changes between self-acupressure with essential oils and sham acupressure and between self-acupressure and sham acupressure from baseline to post intervention were -2.14 and 0.87, respectively, but results were not statistically significant (P = 0.378, P = 0.671).
6	Hughes et al.(2015).[27] UK	To provide preliminary data on the effectiveness of auricular therapy and self-acupressure as compared to standard care in the relief of insomnia in a sample of patients with cancer	Feasibility RCT	N = 7 (IA = 4, 1B = 1, C = 2), patients with breast, prostate or colorectal cancer one to three months post-treatment Mean age = 55.0 yrs, female = 85.7%, time since diagnosis and cancer stage not mentioned	IA: Auricular therapy IB. Self-acupressure	Standard care	Baseline, postintervention (five weeks) Insomnia (PSQI)	Auricular therapy and self-acupressure reported showed an improved trend in insomnia (-4.5 and -4, respectively) from baseline to postintervention, while those in standard care reported no change.

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Table 1
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No.	Author/ Country/ Pub Year	Study Aim	Study Design	Participants	Intervention Group	Control Group	Outcome Measures and Assessment Time Points	Outcome Data
7	Zick et al. (2016).[28] USA	To investigate if six weeks of two types of self-administered acupressure improved fatigue, sleep, and quality of life vs. usual care in breast cancer survivors and to determine if changes were sustained during a four-week washout period	RCT	N = 288 (IA = 98, IB = 94, C = 96), breast cancer survivors with persistent fatigue (≥4 on BFI), cancer-free and at least 12 months post-treatment Mean age = 59.7 to 61.0 yrs, female = 100%, time since diagnosis = 5−5.4 yrs, stage 1 −III (stage I = 30.56%)	I: Relaxing acupressure	C1. Stimulating acupressure C2: Usual care	Baseline, three weeks, postintervention(six weeks), and four-week postintervention (10 weeks) Fatigue (BFI) Sleep quality (PSQI) QOL (Long-term Quality of life Scale) Adverse effects(Self-reported)	Significant greater reductions in fatigue were observed for relaxing acupressure (-2.6 ± 1.5 and -2.3 ± 1.4) and stimulating acupressure (-2.0 ± 1.5 and -2.0 ± 1.5) compared usual care (-1.1 ± 1.6 and -1.0 ± 1.5). at postintervention and 4-week postintervention (both $P < 0.001$), although at neither time were the two acupressure intervention significantly different ($P = 0.29$ and $P > 0.99$). The significant between-group results remained at four-week postinterventions (both $P < 0.001$). For sleep quality, a significant improvement was only observed in the relaxing acupressure group as compared with the usual care, with a change from 8.3 ± 3.8 at baseline to 6.3 ± 2.9 at postintervention (P value not provided). Women in the relaxing group also had a significant improvement in QOL at four-week postintervention as compared to usual care, (P value not provided). Adverse effects occurred in six cases, mainly relating to mild bruising at acupressure sites.

(*Continued*)

Table 1
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No.	Author/ Country/ Pub Year	Study Aim	Study Design	Participants	Intervention Group	Control Group	Outcome Measures and Assessment Time Points	Outcome Data
8	Cheung et al.(2020)[29] Hong Kong	To evaluate the feasibility and potential effects of patient-centered self-administered acupressure for alleviating fatigue and co-occurring symptoms among Chinese advanced cancer patients receiving treatment	Pilot RCT	N = 30 (I = 15, C = 15), advanced cancer patients receiving palliative treatments and with fatigue (≥4 on the "fatigue worst" item of the BFI), and insomnia or pain or both (≥3 on a 0−10 Likert scale) Mean age = 58.9−61.8 yrs, female = 80%, time since diagnosis = 32.4 to 45.7 months, stage III−IV	Self-acupressure	Health education	Baseline, postintervention (after four weeks), four-week follow-up (after eight weeks) Fatigue (BFI) Sleep disturbance (PSQI) Pain (Brief Pain Inventory) Anxiety and depression (HADS) QOL(FACT-G) Adverse effects (Self-reported)	Fatigue reduced in both self-acupressure and control groups at postintervention, with a difference of -0.81 and -0.32, respectively. The between-group difference was -0.83 at postintervention and -0.90 at four-week follow-up, which did not reach statistical significance ($P = 0.83$). For sleep disturbance, the between −group differences at two time points were 2.66 and 0.68, respectively and no statistically significant difference was found ($P = 0.06$).For anxiety, there were a between-group difference of -3.39 and -2.39 at two time points. For depression, the between-group differences were 0.39 and -1.58, respectively. For the severity of pain, the intervention and control groups had a difference of -0.41 and -0.57 at postintervention and four-week postintervention, respectively. For QOL, the differences between two groups are 4.23 (95% -5.22 −13.69) and 1.56 (-10.78−13.91) at postintervention and four-week post-intervention. All of the results were not significant ($P = 0.06$ to 0.84). Adverse effect: None
9	Dogan et al.(2020)[30] Turkey	To determine the effects of acupressure on quality of life and dyspnea level for individuals with lung cancer	RCT	N = 60 (I = 29, C = 31), lung cancer patients undergoing chemotherapy and had experienced moderate to severe dyspnea (>5 on a 0 −10 Likert scale of the Modified Borg Dyspnea Scale) Mean age = 59.0 to 63.1 yrs, female = 11.7%, stages II−IV (stage IV = 63.3%); newly diagnosed but exact time since diagnosis not mentioned	Self-acupressure or by caregivers	Standard care	Baseline and postintervention (after week four) Dyspnea (the modified Borg Dyspnea scale); QOL (St George's Respiratory Questionnaire)	Compared with control group, the intervention group reported a reduction in dyspnea from a median of five to a median of two at baseline and postintervention and the result was significant a ($P = 0.004$). There was also a significant between-group difference in the improvement of QOL ($P < 0.001$) at-postintervention, with a within-group change from 57.71 ± 7.07 to 36.14 ± 9.71 in the intervention group and from 53.83 ± 9.06 to 67.83 ± 9.59 in standard care.

(*Continued*)

Table 1
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No.	Author/ Country/ Pub Year	Study Aim	Study Design	Participants	Intervention Group	Control Group	Outcome Measures and Assessment Time Points	Outcome Data
10	Hoang et al.(2021).[31] Vietnam	To assess feasibility and acceptability of self-acupressure to manage symptom cluster of insomnia, depression and anxiety and to explore its potential effectiveness on severity of symptom cluster, individual symptoms and participants' quality of life	Pilot RCT	N = 114 (I = 38, C1 = 38, C2 = 38), patients with different cancer types during chemotherapy who had insomnia (≥11 on Insomnia Severity Level); Anxiety and Depression (≥8 HADS) Mean age = 55 yrs, female = 72%, stage I−IV (stage IV 53%), mean time since diagnosis = 10.1 months	Self-acupressure	C1: Sham self-acupressure C2: Enhanced standard care	Baseline (before randomization), postintervention (end of week four), one month postintervention (week eight) Insomnia (Insomnia Severity Level) Anxiety and depression (HADS) QOL (FACT-G) Adverse effects (Self-reported)	Compared with enhanced standard care, self-acupressure significantly reduced insomnia at post intervention ($P = 0.002$) and 1-month follow-up ($P = 0.006$), with a change from 20.2 to 15.08 at postintervention to 14.71 at one-month postintervention However, there was no significant difference in insomnia between self-acupressure and sham group ($P = 0.96$ and 0.89), respectively. No significant between-group differences were observed for anxiety ($P = 0.58$ to 0.05), depression ($P = 0.11$ to 0.24) and QOL ($P = 0.67$). Adverse effect: feeling pain at the acupoint after acupressure (8%), feeling finger so tired while stimulating the acupoint (10%)
11	Kim et al (2021)[32] South Korea	To investigate the effects of self-acupressure on chemotherapy-induced peripheral neuropathy, the degree of disturbance in daily activities, and the quality of life in breast cancer patients undergoing chemotherapy	RCT	N = 58 (I = 28, C = 30), breast cancer patients receiving taxane-based chemotherapy and currently experiencing symptoms of peripheral neuropathy due to chemotherapy Mean age = 52.1 yrs, female = 100%, time since diagnosis ≤2 yrs, stage not mentioned	Self-acupressure	Health training	Baseline, postintervention (three weeks) Peripheral neuropathy (European Organization for Research and Treatment of Cancer-Quality of Life Questionnaire Chemotherapy-Induced Peripheral Neuropathy 20) Disturbance in daily life due to peripheral neuropathy (Chemotherapy-induced Peripheral Neuropathy Assessment Tool) QOL (FACT-G)	The intervention group showed a significant decrease in the degree of peripheral neuropathy symptoms (-4.33 vs. 9.4, $P < 0.001$), degree of disturbance in daily activity caused by peripheral neuropathy (-3.46 vs. 13.50, $P = 0.001$) and QOL (4.12 vs. -3.21, $P < 0.001$) compared with the control group.

Note: MFI = Multidimensional Fatigue Inventory, BFI = Brief Fatigue Index; CTCAE = Common Terminology Criteria for Adverse Events; PSQI= Pittsburgh Sleep Quality Index; HADS= Hospital Anxiety and Depression Scale; FACT-G = Functional Assessment of Cancer Therapy—General; QOL = Quality of life

Table 2
Risk of Bias Assessment of the Included RCTs (N = 10)

N.O	Study	Risk of Bias Domain					Overall Risk
		Domain 1: Randomization process	Domain 2: Deviations From the Intended Intervention	Domain 3: Missing Outcome Data	Domain 4: Measurement of the Outcome	Domain 5: Selection of the Reported Result	
1	Dibble et al. (2007)[16]	High	High	Low	High	Low	High
2	Molassiotis et al. (2007)[24]	High	High	Low	Low	Low	High
3	Zick et al. (2011)[25]	Some concerns	Low	Low	Some concerns	Low	Some concerns
4	Tang et al. (2014)[26]	High	High	Low	Low	Low	High
5	Hughes et al. (2015)[27]	Some concerns	High	Low	Some concerns	Low	High
6	Zick et al. (2016)[28]	Some concerns	Low	Low	Some concerns	Low	Some concerns
7	Cheung et al. (2020)[29]	Low	Low	Low	Low	Low	Low
8	Dogan et al. (2020)[30]	Some concerns	High	Low	Some concerns	Low	High
9	Hoang et al. (2021)[31]	Some concerns	Low	Low	Some concerns	Low	Some concerns
10	Kim et al (2021)[32]	High	Some concerns	Low	Low	Low	High

details on randomization or concealment of group allocation, no intention to treat analysis, or not reporting blinding of outcome assessor blinding.

Intervention Characteristics

The intervention characteristics of the included studies, which followed the TIDier checklist, are depicted in Table 3.

Why?. Eleven studies provided the rationale for designing self-acupressure protocol or selecting acupoints, which were informed by previous studies (n = 8),[25-32] expert opinion/consensus (n = 6),[16,25,27,28,30,31] TCM textbook (n = 3),[24,27,31] and practitioner experience (n = 3).[26,29,32]

What Materials?. Most studies (n = 8) provided a printed educational booklet or training guide detailing acupoint locations and acupressure methods associated with symptoms.[16,26−32] Furthermore, eight studies distributed a daily log to check the patients' adherence to acupressure protocol.[17,24-29,32]

What Procedures?. All studies consisted of at least two intervention components, including acupressure skills training by interventionists followed by self-practice. One acupressure skills training session was arranged in nine studies, although details were not often reported.[16,17,24,26,28−32] Four studies arranged one to two extra sessions, ensuring patients could master self-acupressure skills against checklists during and after the study.[26,28,29,31] Five studies included follow-up sessions through telephone calls or face-to-face consultation weekly or biweekly to encourage adherence to acupressure protocol.[16,17,30-32]

Who Provided?. Eleven studies reported the qualifications of interventionists, encompassing acupuncturists,[24,25,27] nurses,[31,32] students with nursing or TCM background,[26,29] and trained research staff.[16,17,28,30] Only two studies provided the training details for non-acupuncturists indicating that each interventionist should complete a 30 or 60 minute training and pass the competency test twice.[28,31]

How?. All studies provided face-to-face acupressure training, eight of which were conducted individually.[16,17,27-32]

Where?. Self-acupressure training was conducted in hospitals or clinics,[16,17,24,27,30-32] whereas self-acupressure was exclusively practiced at home except for two studies.[16,17]

When and How Much?. All studies reported the course of self-practice, ranging from two weeks to five months.[24,26] The number of daily sessions was between one and three depending on symptoms, although once daily was frequently reported.[24,26,28,31] For the intensity of self-acupressure, the stimulating time for each acupoint ranged from 10 seconds to three minutes, with three minutes as the commonly documented.[17,25,28,30,31]

Tailoring. The intervention protocol for all studies was standardized. However, two studies were variant in selecting acupoints, and reported the acupoints selected from a list of fixed acupoints for one particular symptom depending on the patients' condition.[27,29]

Modifications. Modifications were not reported in any studies.

How Well. Three studies documented the patients' fidelity to acupressure training, exceeding 90%.[28,29,31] Five studies reported adherence to intervention protocol at home ranging from 66% to 90.9%.[26,31]

Acupoint Selection. Table 4 presents true and sham acupoint selections for each included study symptom.

True acupoint selection: Among the five studies concerning fatigue, the number of selected acupoints

Table 3

Characteristics of Self-Acupressure Intervention in Included Studies According to the Template for Intervention Description and Replication (TIDieR) Checklist (*n* = 11)

#	Authors	Why	What Materials	What Procedures	Who Provided	How	Where	When and How Much	Tailoring	Modifications	How Well
1	Shin et al. (2004)[16]	Self-acupressure protocol as informed by expert opinion	-Acupressure teaching booklet -Daily log	- One acupressure training (acupoint location, dose, and pressing techniques) - Visit to check the adherence to the protocol -Self-practice before chemotherapy	Research team members trained by experts in East-West Medicine	Individual face-to-face training	Acupressure training, visit and self-practice at hospital	five minutes before chemotherapy, mealtime and anytime sensations of nausea was felt	Standardized	No	Not reported
2	Dibble et al. (2007)[17]	Not reported	Daily log	- One acupressure training (acupoint point and pressing techniques) - One follow-up visit by face-to-face or telephone to encourage the adherence to the protocol -Self-practice before chemotherapy	Research assistant	Individual face-to-face training	Acupressure training, visit & self-practice at hospital	Six minutes in the morning and three minutes each during the rest of the day, depending on the intensity of the nausea; three mins per acupoint	Standardized	No	Not reported
3	Molassiotis et al. (2007)[24]	True acupoints selected to conserve 'energy' informed by TCM meridian theory /textbook	Daily log	- One acupressure training (acupoint locations, dose, and pressing techniques) -Two-week home self-practice at home	One practitioner trained in acupuncture course, with one yr of postqualification clinical experience and previous work experience with cancer patients	Face-to-face training (individual or group training not specified)	-Acupressure training at hospital; -Self-practice at home	Duration of intervention: two weeks seven days per week, once daily, one minute per acupoint	Standardized	No	Daily log completed by all patients
4	Zick et al. (2011)[25]	True acupoint selection to relieve fatigue based on previous literature and expert consensus	-A diagram with written instructions concerning correct pressure and placement of acupressure points -Daily log	-Hand on instructions on acupressure (acupoint location, dose, intensity, and pressing techniques) -Weekly follow-ups by emails or telephone calls to check the need for assistance in locating acupoints and retraining if necessary -12-week self-practice at home	Three acupressure practitioners with two years of clinical experience and with a Master's Degree or above in TCM or related discipline	Face-to-face training (individual or group training not specified)	-Acupressure training at study clinic; -Self-practice at home	Duration of acupressure: 12 weeks seven days per week, two times per day; three minutes per acupoint, in total 27 minutes per time.	Standardized	No	70% adherence rate
5	Tang et al. (2014)[26]	Self-acupressure protocol as suggested by previous literature and practitioner opinion	-A handbook with a colored acupoint map and acupressure methods -Daily log	-One acupressure training on Day one of first chemotherapy (acupoint location, dose, intensity, and pressing techniques) - Confirmed patient self-acupressure accuracy skills on Day two to four and subsequent	Two nursing graduate students trained in acupoint identification and acupressure skills by qualified TCM practitioners after being repeatedly confirmed accuracy of implementation	Face to face training (individual or group training not specified)	-Acupressure training at hospital; -Self-practice at home	Duration of self-acupressure: five months seven days per week, one time per day every morning, each acupoint pressed for one minute, in total six minutes per	Standardized	No	>91.9% adherence rate

(*Continued*)

Table 3
Continued

#	Authors	Why	What Materials	What Procedures	Who Provided	How	Where	When and How Much	Tailoring	Modifications	How Well
				hospitalizations for chemotherapy - Five-month self-practice at home after discharge				application for bilateral sides			
6	Hughes et al. (2015)[27]	Self-acupressure protocol developed based on previous literature, textbooks, expert opinion (with additional considerations to likely ease of location and stimulation by patients)	Daily log	- One 50 minutes acupressure training (acupoint location and application) - Five-week self-practice at home	An acupuncturist with training from British acupuncture society or academy and at least three yrs of clinical experience in treating patients with cancer	Individual face-to-face training	-Acupressure training at hospital; -Self-practice at home	Duration of self-acupressure: five weeks seven days per week, once daily every evening an hour before bedtime, one minute per acupoint	Standardized but acupoints selected for each individual depending on patient condition (one preselected and two self-selected acupoints from five acupoints)	No	One participant completed the logbook as instructed
7	Zick et al. (2016)[28]	True acupoint selection to relieve fatigue based on previous literature and expert consensus	- Written instruction manual - Daily log	- One 15-minutes training session - Two follow-up visits to assess acupressure skills at week three and week six using a checklist -Six-week self-practice at home	13 acupressure educators who received a 30 minutes training by a licensed acupuncturist and passed the competency test twice	Individual face-to-face training and follow-up visits	-Acupressure training and follow-up at country office -Self-practice at home	Duration of self-acupressure: six weeks. seven days per week, once daily, three minutes per acupoint	Standardized	No	Fidelity rate 95.3% Adherence rate 73%
8	Cheung et al. (2020)[29]	Self-acupressure protocol developed based on previous literature and practitioner opinion	- A resource book (relevant theories and the acupressure protocol) - A booklet and a poster for learning - Daily log	- Two-hours acupressure training sessions (acupoint function, location, dose, intensity, and pressing techniques) - Three-weeks self-practice at home -Three 1-hour weekly follow-up visits for reinforcing practice -Postintervention fidelity test for checking the accuracy of self-acupressure	Six trainers including three pairs of a senior-year TCM student and a nursing student, who were trained and certified by two licensed TCM practitioners	Individual face-to-face training and follow-up visits	-Training at the patient's home or at the university; -Self-practice at home	Duration of self-acupressure: three weeks seven days per week, one to two minutes per acupoint, each session 15 minutes, twice every day (once in the morning and once in the afternoon or evening), in total 10.5 hours	Standardized but acupoints selected for each individual according to presenting symptoms (four preselected and two self-selected acupoints as recommended by trainers)	No	Adherence rate 90.9; Competency test: 42.8 (SD = 2.03, maximum 44)
9	Dogan et al. (2020)[30]	The acupoint selection based on previous literature and expert opinion	Acupressure application training guide	-One acupressure training (importance and practice of applying acupressure to the correct acupoints, acupressure intensity and techniques) - Three weekly phone follow-ups to check for continued application of the acupressure - Four week self-practice at home	One acupressure educator by research team member, qualifications and training not specified	Individual face-to-face training and phone follow-ups	-Training at hospital -Self-practice at home	Duration of self-acupressure: four weeks seven days per week, twice every day, three minutes per acupoint, each session 18 minutes, total 36 minutes daily	Standardized	No	Not reported

(*Continued*)

Table 3
Continued

#	Authors	Why	What Materials	What Procedures	Who Provided	How	Where	When and How Much	Tailoring	Modifications	How Well
10	Hoang et al. (2021)[31]	Self-acupressure protocol including acupoint selection developed based on TCM theory, previous review and expert consensus	- Booklet: with images illustrating self-acupressure protocol - Written leaflet: on sleep hygiene measures alongside ways to manage anxiety and depression	- One 30-minutes training including acupressure training session and recommendations of symptom management - Four-week self-practice at home - One 5- 10 minutes follow-up visit at next chemotherapy cycle to check for the location and pressure techniques - Weekly phone calls to remind about practice of self-acupressure	Three nurses who had at least eight yrs experience of working with cancer patients and received a 60-min training session from a TCM doctor.	Individual face-to-face training, follow-up visits, phone calls	-Training at hospital; -Self-practice at home	Duration of self-acupressure: four weeks seven days per week, three minutes per acupoint, 28 minutes per session, once every night 30 −60 minutes before going to bed	Standardized	No	Adherence rate: 66%; fidelity: 93%
11	Kim et al (2021)[32]	The acupoint selection to relieve chemotherapy-induced peripheral neuropathy based on previous literature and practitioner opinion	- Educational booklet - Daily log	- One 30-minutes acupressure training (Acupoint location, and pressing methods) Three-week self-practice at home - Bi-weekly phone calls to check the adherence to the protocol	A oncology nurse working with >3 yrs of clinical experience, but qualification in acupressure not described	Individual face-to-face training and follow-up phone calls	-Training at hospital, - Self-practice at home	Duration of self-acupressure: three weeks seven days per week, 10 seconds per acupoint, three times daily	Standardized	No	Not reported

Table 4

Summary of True and Shame Acupoints in Included Studies Based on the WHO Standard Acupuncture Point Locations in the Western Pacific Region ($n = 11$)

			True Acupoints																						Sham Acupoints				
#	Study ID	Symptom	PC6	ST36	SP6	LI4	HT7	LV3	KI3	CV6	GV20	GB20	KI6	BL62	BL23	LU1	LU10	CV4	LI11	LI10	EX-HN3 #	EX-HN22 #	EX-UE9 e#	EX-LE10#	SI3	LI12	GB33	BL61	Nonacupoint
1	Shin et al. (2004)[16]	Nausea and vomiting	v																										
2	Dibble et al. (2007)[17]	Nausea and vomiting	v																						v				
3	Molassiotis et al. (2007)[24]	Fatigue		v	v	v																				v	v	v	
4	Zick et al. (2011)[25]	Fatigue			v		v	v													v	v							
5	Tang et al. (2014)[26]	Fatigue		v	v	v																							
6	Hughes et al. (2015)[27]	Insomnia	v				v					v	v	v															
7	Zick et al. (2016)[28]	Fatigue			v		v	v													v	v							
8	Cheung et al. (2020)[29]	Fatigue	v	v	v				v	v					v			v											
		Insomnia			v		v				v	v																	
		Pain	v	v	v	v		v	v		v	v																	
		Depression	v	v			v	v			v	v																	
		Anxiety		v			v	v			v	v																	
9	Dogan et al. (2020)[30]	Dyspnea	v													v	v												v
10	Hoang et al. (2021)[31]	Insomnia	v				v				v	v									v								v
		Depression					v	v			v	v									v								v
		Anxiety					v	v			v	v									v								v
11	Kim et al (2021)[32]	Chemotherapy-induced peripheral neuropathy		v		v													v	v			v	v					

PC6 = Pericardium 6 (Neiguan 內關), ST36 = Stomach 36 (Zusanli足三里), SP6 = Spleen 6 (Sanyinjiao三陰交), LI4 = Large intestine (Hegu合谷), HT7 = Heart 7 (Shenmen神門), LV3/ LR3 = Liver 3 (Taichong太衝), KI3 = Kidney 3 (Taixi 太溪), CV6 = Conception vessel 6 (Qihai 氣海), GV20 = Governing vessel 20 (Baihui 百會), GB20 = Gallbladder 20 (Fengchi 風池), KI6 = Kidney 6 (Zhaohai照海), BL62 = Bladder 62 (Shenmai申脈), BL23 = Bladder 23 (Shenshu腎俞), LU1 = Lung 1 (Zhongfu中府), LU10 = Lung 10 (Yuji魚際), CV4 = Conception vessel 4 (Guanyuan關元), LI11 = Large intestine 11 (Quchi 曲池), LI10 = Large intestine 10 (Shousanli手三里)

Acupoints not shown in the WHO standard acupuncture point locations in the Western Pacific Region. EX-HN3 (Yintang印堂), EX-HN22 (Anmian,(安眠), EX-UE9 (Baxie八邪), EX-LE10 (Bafeng 八風).

SI3=Small Intestine (Houxi后溪); LI12= Large intestine 12 (Zhouliao 肘髎), GB33= Gallbladder 33 (Xiyangguan膝陽關), BL61= Bladder 61 (Pucan仆參)

ranged from three to seven.[24,29] The most commonly used acupoints were SP6 (Sanyinjiao) and ST36 (Zusanli). Among the three studies concerning insomnia, four to six acupoints were chosen.[29,31] HT7 (Shenmen) and GB20 (Fengchi) were used in three studies, whereas GV20 (Baihui) was applied in two studies. For two studies regarding depression and anxiety, the number of selected acupoints was five to six, with the most consistently reported acupoints including HT7(Shenmen), GV20 (Baihui), and GB20 (Fengchi).[29,31] For nausea and vomiting, two studies applied on a single acupoint only (PC6, Neiguan).[16,17]

Sham acupoint selection: Among the four studies that reported the use of sham acupoints.[17,24,30,31] two adopted true acupoints that were not associated with symptoms,[17,24] and the remaining four used nonacupoints, which were some distance from true acupoints.[30,31]

Outcomes

Fatigue Four feasibility/pilot trials and one RCT assessed fatigue using different instruments.[24-26,28,29] An earlier feasibility trial measured fatigue using the Multidimensional Fatigue Inventory and found that self-acupressure was significantly more effective than the sham group in reducing fatigue, prominently for general fatigue ($P = 0.024$ to <0.001).[24] The reductions in general fatigue for self-acupressure and sham groups from baseline to postinterention were 3.2 and 0.1, respectively. A pilot RCT found a mean difference of 17.66 in the change of fatigue as measured by the Tang Fatigue Rating scale between self-acupressure and sham control but did not reach a statistically significant level ($P = 0.59$).[26] Among the two pilot RCTs and one RCT assessing fatigue using the Brief Fatigue Inventory,[25,28,29] only a RCT demonstrated a significant between-group difference in fatigue between self-acupressure and usual care ($P < 0.001$), with a mean difference of 1.5.[28]

Insomnia: The effect of self-acupressure on insomnia was reported in five RCTs.[26-29,31] Three feasibility/pilot RCTs and one RCT assessed insomnia using the Pittsburgh Sleep Quality Index. They documented an improvement of 0.45 to four from baseline to postintervention in the self-acupressure groups.[26-29] Among the four trials, one pilot RCT found a statistically significant difference of 2.25 in insomnia between self-acupressure and sham control ($P = 0.04$).[26] In contrast, another RCT observed a significant improvement in insomnia in the relaxing acupressure group compared to usual care at postintervention (actual values not provided).[28] Another pilot RCT measured insomnia using the Insomnia Severity Level, and the result showed that compared with enhanced standard care, self-acupressure significantly reduced insomnia with a reduction from 20.2 to 15.08 at postintervention ($P = 0.002$).

Anxiety and depression: The effects of acupressure on anxiety and depression using the Hospital Anxiety and Depression Scale were examined in three pilot RCTs.[26,29,31] However, none of the trials identified a significant between-group difference in anxiety ($P = 0.11-79$) or depression ($P = 0.10 -0.24$).[29,31]

Nausea and vomiting: One RCT and one quasi-experimental study measured nausea and vomiting using the Rhodes Index of Nausea, Vomiting and Retching and consistently found a significant between-group difference in the intensity of nausea and vomiting ($P = 0.006$ and $P = 0.001$), respectively.[16,17] The quasi-experimental study reported a mean difference of 3.6 between self-acupressure and usual care.[16]

Other symptoms: Three RCTs examined the effects of self-acupressure on pain, dyspnea, and chemotherapy-induced peripheral neuropathy[29,30,32] separately. For the severity of pain as measured by the Brief Pain Inventory, the pilot study found a difference of -0.41 and -0.57 between self-acupressure and health education at postintervention and four-week follow-up, respectively, but the results were not statistically significant ($P = 0.84$).[29] Another RCT measured dyspnea using the modified Borg Dyspnea Scale and found that compared with usual care, the self-acupressure group had a significant reduction in dyspnea with a median difference of three ($P = 0.004$). One RCT measured chemotherapy-induced peripheral neuropathy using the European Organization for Research and Treatment of Cancer-Quality of Life Questionnaire Chemotherapy-Induced Peripheral Neuropathy 20-item scale and found that the intervention group showed a significant decrease in chemotherapy-induced peripheral neuropathy as compared with the control group (-4.33 vs. 9.4, $P = 0.001$).[32]

QOL: Five RCTs assessed QOL using the Functional Assessment of Cancer Therapy-General,[29,31,32] Long-term Quality of life Scale,[28] and St George's Respiratory Questionnaire,[30] respectively. Three RCTs consistently indicated that self-acupressure improved QOL at post-intervention or four-week follow-up ($P < 0.001$).[28,30,32] By contrast, two pilot RCTs reported no significant between-group difference in QOL($P = 0.06-0.667$).[29,31]

Adverse Effects

Among the five trials that reported adverse effects of self-acupressure,[24,25,28,29,31] only four identified mild issues among a few cases, including bruising or pain at the acupoints,[24,28] finger tiredness,[31] musculoskeletal, dizziness, hot flashes, and sleep issues.[25] The remaining studies did not report whether adverse effects took place ($n = 6$).

Discussion

Eleven studies, predominantly feasibility or pilot RCTs, were included in the analysis after a systematic review of the current experimental research on self-acupressure for cancer symptom management. The results of the analysis found that self-acupressure was associated with reduced nausea and vomiting, which is in agreement with a similar review but with focus on therapist-administered acupressure.[6] However, the effects of self-acupressure on other cancer symptoms including fatigue, insomnia, anxiety and depression, dyspnea, and chemotherapy-induced peripheral neuropathy remained inconclusive, which might be largely explained by the small sample size and the small number of studies for each symptom.

Although the included studies contained similar core components and delivery of self-acupressure including face-to-face skills training by interventionists supplemented with educational materials at hospitals/clinics and self-practice at home, they were heterogeneous in terms of the qualifications of interventionists, acupoint selection, duration, timing, and intensity of the intervention. Evidence shows these characteristics of self-acupressure determine the effect on symptoms through accurate pressuring method and dosage to achieve a "*de qi*" sensation [33,34] In view of self-acupressure as a self-management technique, the extent to which the intervention was executed by patients as planned is the key to success.[35,36] According to treatment fidelity recommendations by the National Institutes of Health Behavior Change Consortium,[37] five parameters of fidelity should be considered, encompassing study design, training providers, intervention delivery, receipt of intervention, and enactment of intervention skills. In this review, these aspects were inadequately reported or not reported at all. Although all studies included acupressure skills training, the content and time of training was seldom fully described, which made it difficult to determine whether adequate training was received by patients. Furthermore, few studies ensured fidelity by arranging follow-up sessions to check the patients' skill mastery. In addition, most studies used a daily log to check the intervention adherence at home, but actual receipt of intervention was only reported in a few studies. Notably, enhancing self-acupressure technique alone may be not adequate to achieve optimal adherence to the intervention in the short and long term because other factors may play an important role, for example, belief in TCM, establishing a habit of self-acupressure, perceived health benefits of self-acupressure, and availability of family support.[36] Hence, to promote the duplication of future research, adequate measurement and reporting of fidelity to self-acupressure is necessary.

The quality of the included studies was almost poor mainly because of incomplete or poor reporting of randomization, concealment, blinding, and intention to treat analysis. Most trials included in this review were on a small scale, and had no randomization, concealment, and blinding, which could be due to manpower issues. However, a methodologically rigorous trial with self-acupressure for cancer symptoms is significantly challenging in real clinical settings. Hence, achieving the intervention efficacy through conducting pragmatic trials for self-acupressure may be considered in future research.[38,39]

Limitations

This review has several limitations. First, language constraint was a limitation because of the possibility of excluding studies published in non-English or non-Chinese languages. Second, the majority of trials were exploratory in nature, rendering the interpretation of the effectiveness of self-acupressure being treated with caution. Third, the review was limited by variability in the intervention protocol and control groups, making cross-study comparison difficult. Lastly, the included trials was inferior in methodological quality, further adding to the uncertainty of true effectiveness of intervention.

Implications

The studies evaluating the role of self-acupressure for cancer symptoms included in this review are scant and self-acupressure merits further evaluation in the context of cancer care. Despite limited evidence, this review identified the possible beneficial effects of self-acupressure on reducing nausea and vomiting in cancer patients. However, because of the absence of the minimum clinically important difference (MCID) value for the Rhodes Index of Nausea, vomiting and retching, clinical significance is not determined. The use of MCIDs to aid the interpretation of clinical trials involving patient-reported outcomes has been increasingly used to guide accurate conclusions in order to affect clinical practice.[40] In addition, some studies in this review did not provide actual values for within-and between group differences, making it difficult to capture the trend of the effectiveness of self-acupressure for symptoms over time. Thus, future research is encouraged to report the magnitude of data across interventions and time points. Furthermore, the reporting of self-acupressure intervention characteristics in individual studies according to the TIDieR checklist was generally complete, although specific descriptions of intervention fidelity were lacking. Future studies are warranted to follow a more comprehensive approach to reporting of intervention fidelity based on recommendations by the National Institutes of Health Behavior Change Consortium to promote the quality of self-acupressure trial.[35,37] A recent review documented the increasing use of smartphone applications to support self-acupressure by monitoring competence and adherence to the intervention via self-assessment tools, timers, reminders, and schedule checkers.[41] The use of information and communication technologies may be a possible strategy for promoting the application of self-acupressure for cancer symptom management in post-COVID era.

Conclusion

Self-acupressure remains an emerging intervention for cancer symptom management, and hence, the existing experimental evidence is predominantly explorative in nature. Despite the limited studies, self-acupressure may be a potentially effective self-management option for cancer patients with nausea and vomiting. However, the small number of studies, inadequate sample sizes, lack of consistent rigorous methodology, and the heterogeneity in intervention protocols and control groups across the studies preclude the definitive conclusions on intervention effectiveness for the studied symptoms. Future research should consider developing standard protocols for intervention delivery, improving the methodology of self-acupressure, and conducting large-scale research to advance the science of self-acupressure for cancer symptom management.

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Appendix I
PRISMA 2020 Checklist

Section and Topic	Item #	Checklist Item	Location Where Item is Reported
TITLE			
Title	1	Identify the report as a systematic review.	1
ABSTRACT			
Abstract	2	See the PRISMA 2020 for Abstracts checklist.	1
INTRODUCTION			
Rationale	3	Describe the rationale for the review in the context of existing knowledge.	1–2
Objectives	4	Provide an explicit statement of the objective(s) or question(s) the review addresses.	2
METHODS			
Eligibility criteria	5	Specify the inclusion and exclusion criteria for the review and how studies were grouped for the syntheses.	2
Information sources	6	Specify all databases, registers, websites, organizations, reference lists and other sources searched or consulted to identify studies. Specify the date when each source was last searched or consulted.	2
Search strategy	7	Present the full search strategies for all databases, registers and websites, including any filters and limits used.	Appendix III
Selection process	8	Specify the methods used to decide whether a study met the inclusion criteria of the review, including how many reviewers screened each record and each report retrieved, whether they worked independently, and if applicable, details of automation tools used in the process.	2
Data collection process	9	Specify the methods used to collect data from reports, including how many reviewers collected data from each report, whether they worked independently, any processes for obtaining or confirming data from study investigators, and if applicable, details of automation tools used in the process.	2
Data items	10a	List and define all outcomes for which data were sought. Specify whether all results that were compatible with each outcome domain in each study were sought (e.g., for all measures, time points, analyses), and if not, the methods used to decide which results to collect.	2
	10b	List and define all other variables for which data were sought (e.g., participant and intervention characteristics, funding sources). Describe any assumptions made about any missing or unclear information.	2
Study risk of bias assessment	11	Specify the methods used to assess risk of bias in the included studies, including details of the tool(s) used, how many reviewers assessed each study and whether they worked independently, and if applicable, details of automation tools used in the process.	3
Effect measures	12	Specify for each outcome the effect measure(s) (e.g., risk ratio, mean difference) used in the synthesis or presentation of results.	3
Synthesis methods	13a	Describe the processes used to decide which studies were eligible for each synthesis (e. g., tabulating the study intervention characteristics and comparing against the planned groups for each synthesis (item #5)).	3
	13b	Describe any methods required to prepare the data for presentation or synthesis, such as handling of missing summary statistics, or data conversions.	3
	13c	Describe any methods used to tabulate or visually display results of individual studies and syntheses.	3
	13d	Describe any methods used to synthesize results and provide a rationale for the choice (s). If meta-analysis was performed, describe the model(s), method(s) to identify the presence and extent of statistical heterogeneity, and software package(s) used.	3
	13e	Describe any methods used to explore possible causes of heterogeneity among study results (e.g., subgroup analysis, meta-regression).	N/A
	13f	Describe any sensitivity analyses conducted to assess robustness of the synthesized results.	N/A
Reporting bias assessment	14	Describe any methods used to assess risk of bias due to missing results in a synthesis (arising from reporting biases).	N/A
Certainty assessment	15	Describe any methods used to assess certainty (or confidence) in the body of evidence for an outcome.	N/A
RESULTS			
Study selection	16a	Describe the results of the search and selection process, from the number of records identified in the search to the number of studies included in the review, ideally using a flow diagram.	3
	16b	Cite studies that might appear to meet the inclusion criteria, but which were excluded, and explain why they were excluded.	3
Study characteristics	17	Cite each included study and present its characteristics.	3
Risk of bias in studies	18	Present assessments of risk of bias for each included study.	9
Results of individual studies	19	For all outcomes, present, for each study: 1) summary statistics for each group (where appropriate) and 2) an effect estimates and its precision (e.g., confidence/credible interval), ideally using structured tables or plots.	4-8
Results of syntheses	20a	For each synthesis, briefly summarize the characteristics and risk of bias among contributing studies.	3,9.14

(*Continued*)

Appendix I
Continued

Section and Topic	Item #	Checklist Item	Location Where Item is Reported
	20b	Present results of all statistical syntheses conducted. If meta-analysis was done, present for each the summary estimate and its precision (e.g., confidence/credible interval) and measures of statistical heterogeneity. If comparing groups, describe the direction of the effect.	14
	20c	Present results of all investigations of possible causes of heterogeneity among study results.	N/A
	20d	Present results of all sensitivity analyses conducted to assess the robustness of the synthesized results.	N/A
Reporting biases	21	Present assessments of risk of bias due to missing results (arising from reporting biases) for each synthesis assessed.	N/A
Certainty of evidence	22	Present assessments of certainty (or confidence) in the body of evidence for each outcome assessed.	N/A
DISCUSSION			
Discussion	23a	Provide a general interpretation of the results in the context of other evidence.	15
	23b	Discuss any limitations of the evidence included in the review.	15
	23c	Discuss any limitations of the review processes used.	15
	23d	Discuss implications of the results for practice, policy, and future research.	15-16
OTHER INFORMATION			
Registration and protocol	24a	Provide registration information for the review, including register name and registration number, or state that the review was not registered.	3−4
	24b	Indicate where the review protocol can be accessed, or state that a protocol was not prepared.	3−4
	24c	Describe and explain any amendments to information provided at registration or in the protocol.	N/A
Support	25	Describe sources of financial or nonfinancial support for the review, and the role of the funders or sponsors in the review.	Title page
Competing interests	26	Declare any competing interests of review authors.	Title page
Availability of data, code and other materials	27	Report which of the following are publicly available and where they can be found: template data collection forms; data extracted from included studies; data used for all analyses; analytic code; any other materials used in the review.	N/A

Appendix II
Key Search Terms

	Population	Concept
English	cancer OR tumor OR tumour OR carcinoma OR oncolog* OR neoplas*	Self-acupressure or self-administered acupressure
Chinese	癌 OR 肿瘤	自我穴位按压

Appendix III
Full Electronic Search Strategy (Search Date: January 29, 2022)

Database	Search strategy	Result
PubMed	1. "neoplasms"[MeSH Terms] OR "carcinoma"[MeSH Terms]	361,316
	2."Cancer"[Title/Abstract] OR "tumor"[Title/Abstract] OR "tumour"[Title/Abstract] OR "carcinoma"[Title/Abstract] OR "oncolog*"[Title/Abstract] OR "neoplas*"[Title/Abstract]	3,324,481
	3. “Self-acupressure” [Title/Abstract]OR “self-administered acupressure” [Title/Abstract]	59
	4. 1 OR 2	4,578,996
	5. 4 AND 3	15
CINAHL Complete (via EbscoHost)	1. (MH "Neoplasms") OR (MH "Carcinoma+")	97,434
	2. TI (Cancer OR tumor OR tumour OR carcinoma OR oncolog* OR neoplas*) OR AB (Cancer OR tumor OR tumour OR carcinoma OR oncolog* OR neoplas*)	622,885
	3. TI (Self-acupressure OR self-administered acupressure) OR AB (Self-acupressure OR self-administered acupressure)	64
	4. 1 OR 2	643,510
	5. 4 AND 3	14
British Nursing Index	1. ab(Cancer OR tumor OR tumour OR carcinoma OR oncolog* OR neoplas*) OR ti(Cancer OR tumor OR tumour OR carcinoma OR oncolog* OR neoplas*)	65,639
	2. TI (Self-acupressure OR self-administered acupressure) OR AB (Self-acupressure OR self-administered acupressure)	10
	3. S1 & S2	2
Cochrane	1. MeSH descriptor: [Neoplasms] explode all trees	85,747
	2. MeSH descriptor: [Carcinoma] explode all trees	14,432
	3. (Cancer OR tumor OR tumour OR carcinoma OR oncolog* OR neoplas*):ti OR (Cancer OR tumor OR tumour OR carcinoma OR oncolog* OR neoplas*):ab	207,437
	4. (Self-acupressure OR self-administered acupressure):ti OR(Self-acupressure OR self-administered acupressure):ab	2,442
	5. #1 OR #2 OR #3	227,728
	6. #5 AND #4	291
	7.# trials	228
EMBASE	1. ’neoplasms’/exp OR ’carcinoma’/exp	5,442,742
	2. cancer:ab,ti OR tumor:ab,ti OR tumour:ab,ti OR carcinoma:ab,ti OR oncolog*:ab,ti OR neoplas*:ab,ti	4,454,355
	3. Self-acupressure:ab, ti OR self-administered acupressure:ab,ti	114
	4. #1 OR #2	6,290,966
	5. # 4 AND # 3	36
Web of Science	1. TI=(Cancer OR tumor OR tumour OR carcinoma OR oncolog* OR neoplas*) OR AB=(Cancer OR tumor OR tumour OR carcinoma OR oncolog* OR neoplas*)	3,651,314
	2. TI=(Self-acupressure) OR AB=(self-administered acupressure)	43
	3. 1 AND 2	12
CNKI	1. (篇名:癌 + 肿瘤) OR (摘要:癌 + 肿瘤)	5,842,854
	2. (篇名:自我穴位按压) OR (自我穴位按压) (结果中检索)	65
Index to Taiwan Periodical Literature System	1. (TI=癌 OR 腫瘤) [OR] (IT=自我穴位按压)	0
	2. (AB=癌 OR 腫瘤) [AND] (AB=自我穴位按压)	0
	3. 1 OR 2	0